Complete design and firmware review

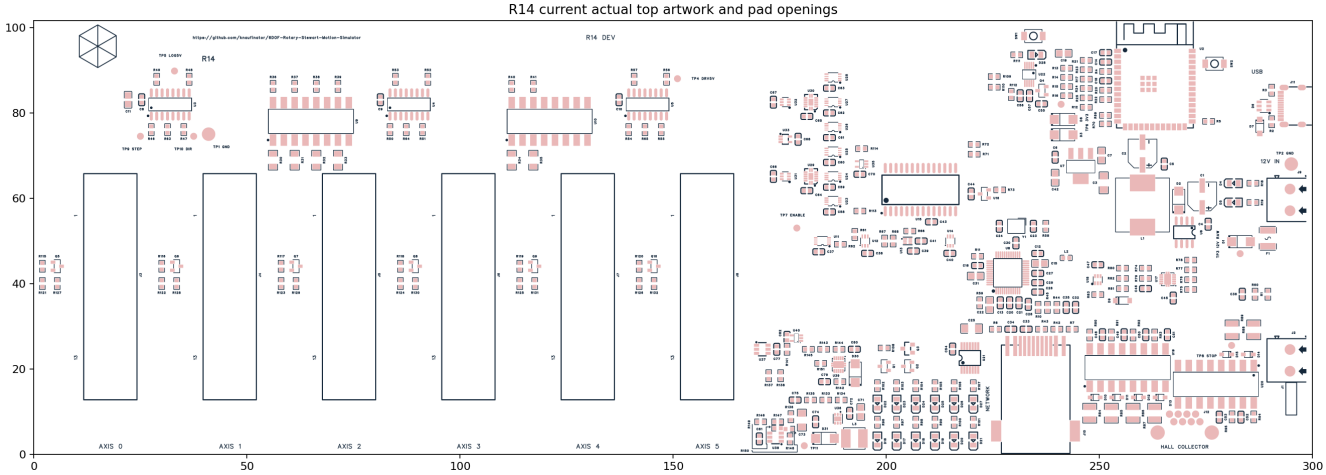
25 October 2026 | Current EasyEDA master + current repository

RELEASE STATUS: HOLD

The digital PCB checks pass. The present repository does not contain the R13 firmware implementation that produced the latest successful bench results. R14 also requires a live peripheral/enable/homing adapter, and the field-supply, power-sequence, assembly and physical-interface qualifications remain open.

This report preserves the working R13 evidence as the required baseline. It does not certify that the current checkout reproduces that behavior or that an assembled R14 has been tested.

R14 current actual top artwork and pad openings



Reviewed coverage	Result
319 components / 1,195 logical pins	Identity and pin-net agreement PASS
1,724 traces / 878 vias / six pours	Actual-CAM structural connectivity PASS
355 designator, text and digit records	Identity and artwork checks PASS
90 native BOM groups / 319 CPL references	BOM/CPL consistency PASS
Working R13 changes / R14 feature firmware	Not preserved / not implemented in this checkout

No PCB or firmware source was changed. No flashing, motor motion, desktop control, order, commit or push was performed. Review binaries are build evidence only.

Read this first

Decision and meaning of the results

Do not treat this revision as a production-ready system. Two P0 findings prevent a firmware compatibility sign-off: the measured R13 pulse-engine replacement is missing, and the R14 hardware has no live qualified firmware adapter. The default current firmware deliberately blocks normal motion. A successful build or JLC import does not close either finding.

The highest-priority hardware margin issue is FIELD_VGOOD: its nominal local threshold is 9.522 V, below the selected sensor's 10 V minimum even before the retained 15 ft cable loss. Existing power-startup and factory-process qualifications also remain conditional. These are documented findings, not new source changes made during this review.

What was checked exhaustively

All 319 source components were cross-checked for purchasing identity and logical pin-net agreement between schematic and PCB. Every native copper trace/via/fill/test-pad object has a register entry tied to actual plotted copper. Gerber polarity and drilled holes were included, and layers were joined only through real plated connections. Every current designator, board string and fixed DB25 numbering digit has an entry. Artwork tiles and the corrected connector digits were visually inspected.

What this does not certify

Per-component PASS in the register means identity and connectivity. It does not mean every tolerance, absolute maximum, thermal state, transient, analog stability condition or assembly yield has been proven. Reference-circuit and datasheet comparisons are recorded by section, with unresolved margins stated explicitly. No new physical R14 test, complete switching simulation, factory allocation review or full assembled-STEP/enclosure qualification was performed. This is a complete digital design review with named open qualifications, not a claim that all real-world conditions pass.

Priority definitions

Priority	Meaning
P0	Blocks compatibility or feature-completeness sign-off.
P1	Must be resolved or explicitly accepted for a bounded prototype decision.
P2	Integration, documentation or performance qualification work remains.

Navigation

The report contains the source baseline, 13 findings, 20 section reviews, the R13 feature-preservation matrix, testpoint contract, validation evidence, references, and an all-parts appendix. The accompanying CSV files provide every pin, copper primitive and label without shrinking thousands of rows into unreadable PDF pages.

Frozen review baseline

Item	Reviewed identity
EasyEDA project	R14 G1SOL7C4 BOOST ROUTED TRIAL DO NOT ORDER 20261023
Project UUID	df6039acdfad4375ae07aa6257cd0186
PCB UUID	6567ed8798fb578fbfd0297236c2ee0e
Schematic page UUID	2d5089592f955b849d84cabed91b1f32
Repository branch	r13-single-motor-sim-20261011
Repository HEAD	7cc29d43fafb2de6d7da733621495c489b70ed64
Shared core	118e6b0965cee674994bd66d7b9fb7f07c7420e8
Working tree	Pre-existing dirty documentation/app changes captured in repo-status.txt and repo-diff.patch; not discarded or revised.

after-project.epro2

SHA-256 d8311057b5f2ab4497c16169ba91a2d3b68f73e026eec1959fb1150e0d3c352a

current-Gerber-FS36.zip

SHA-256 f79201957e2f3f11c8bf36bad75b3013b57652ab0a9987926484725b39b4e1fa

The current archive has no non-header record difference from the saved grouped master. Raw API source and portable archives use different serializers, so a raw text comparison is not a geometry certificate. Their differing precision/aliases/rule serialization is retained in the comparison receipts; archive-to-archive identity, source/net agreement and actual CAM are the checks used here.

Earlier authorized master changes

The intentionally unpopulated C78 filter was removed from schematic and PCB. Value/Comment fields were standardized to group identical supplier parts in the native BOM. Current supplier identities, retained geometry and net memberships agree, with 319 required references. This review evaluates that resulting master; it does not reintroduce C78 or omit fitted components to clear warnings.

R13 evidence authority

Latest dated authority: October 20 BRINGUP_LOG, pulse-accounting handoff and USB DTR handoff. These describe successful physical bench runs and the exact uncommitted working tree on another workstation at C:/GitHub/6dof-r13-bench. That working tree is absent here. The earlier host-streamed heave run was explicitly retracted because targets were computed without motion; it is not counted as a pass. The later on-controller waveform/soak evidence remains valid historical evidence.

Findings overview

ID	Priority	Finding
F01	P0	Tested R13 pulse-engine fixes are absent from this checkout
F02	P0	R14 feature hardware has no live qualified firmware adapter
F03	P1	Latest measured 2:1 gearing and angular limits are not preserved
F04	P1	Seek contract still requires an obsolete physical hold condition
F05	P1	All-axis alarm command stop is not a live implemented path
F06	P1	R14 cable and drive-enable contract differs from the working R13 setup
F07	P1	FIELD_VGOOD threshold does not certify the selected remote sensor voltage
F08	P1	Auxiliary boost and power-sequence guarantees remain conditional
F09	P1	Assembler process approval is still required for reduced margins
F10	P1	Collector wrong-port protection and actual mechanical mating remain open
F11	P2	USB host-line-state and GPIO readback fixes are not in this repo
F12	P2	Old docs and source comments can select the wrong baseline
F13	P2	Ethernet/SPI/USB connectivity pass is not complete signal-integrity qualification

Open does not always mean a demonstrated electrical failure. Findings distinguish confirmed source mismatches from qualifications for which the review lacks sufficient physical or process evidence.

F01 Tested R13 pulse-engine fixes are absent from this checkout

P0 | CONFIRMED SOURCE REGRESSION

The October 20 bench handoff identifies three production-engine defects and a tested replacement using TEP-latched force gates, direct level-3 interrupt handling and phase/latch accounting. Current source still arms an immediate compare value, resets it to zero in the same ISR when a target is reached, registers on_full callbacks, and enables HIGH-on-TEZ. These are the structures the measured R13 fixes replaced. The bench handoff explicitly locates the working, uncommitted source at C:/GitHub/6dof-r13-bench on a different workstation. That checkout is not present here.

Required closure: Recover and review the exact tested working tree, including its CMake flags and submodule changes; preserve its source/binary hashes. Port that implementation before enabling R14 motion and replay pad-counter/camera tests at 1, 5, 20, 100 and 250 kHz, including final-step and reversal cases.

Evidence: controller/include/SharedMcpwmStepEngine.h:235; controller/include/SharedMcpwmStepEngine.h:465; G:/My Drive/projects/6dof/pcb/HANDOFF_20261020_STEP_ENGINE_PULSE_ACCOUNTING.md

F02 R14 feature hardware has no live qualified firmware adapter

P0 | CONFIRMED MISSING IMPLEMENTATION

The current ESP32-S3 build deliberately constructs MotionAdmission<false>. normal_motion_permitted() always returns false; the alarm callback has no live GPIO producer. There is no full-size MCP23017/INA226 driver or S3 enable initialization/arm transaction in this checkout. The R14 Hall, field-power, latch and ADC features therefore cannot be claimed implemented by either successful compilation or JLC import. R13_BENCH_ONE_AXIS is a separate console-only axis-0 route.

Required closure: Implement the R14 adapter and tested initialization/recovery state machine after F01. Configure MCP23017 GPA7/GPB7 as outputs, establish low arm clock before releasing clear, sample all six NC/alarm channels, read field/enable/ADC states, and retain fail-closed on stale I2C or unqualified inputs.

Evidence: controller/main/main.cpp:180; controller/include/R14MotionAdmission.h:90; controller/main/CMakeLists.txt:88

F03 Latest measured 2:1 gearing and angular limits are not preserved

P1 | CONFIRMED SOURCE / BENCH MISMATCH

The latest R13 log records Pn098=2, Pn102=1, 694.4445 pulses per arm degree and a 50000-step 72-degree limit. Current shared-core defaults remain virtual_gear=80 and main.cpp sets fixed +/-100000-step limits. DRIVE setters have no reviewed finite/range validation, persistence or limit reapplication. At a physical gear of 2, 100000 steps represents 144 degrees; a gear-80 software default commands 1/40 of the intended movement. This is a source mismatch, not a claim about the currently flashed board.

Required closure: Recover the bench-tested gear=2 changes and persistent validated drive configuration; recompute measured per-axis angular limits on every configuration change. Read back actual drive settings before commissioning. Keep Pn051=3000 as the retained owner setting.

Evidence: controller/stewart-core/src/InverseKinematics.cpp:140; controller/main/main.cpp:500; controller/main/main.cpp:1814; G:/My Drive/projects/6dof/pcb/r13-bringup-notes-20261012/BRINGUP_LOG.md

F04 Seek contract still requires an obsolete physical hold condition

P1 | CONFIRMED REQUIREMENT MISMATCH

The later owner decision requires explicit arming and commanded motion without a physical hold-to-run button. SeekInterlock still requires local_hold continuously and a release-to-press edge. This cannot be solved by treating an unconnected input as permanently true: the class also requires a fresh edge and is the basis of the host tests. LOCAL_HOLD_A/B remain available hardware nets, but presence does not reinstate the superseded requirement.

Required closure: Revise the admission contract to represent deliberate operator authorization and separate explicit home request, preserving NC Hall presence, bounded seek, alarm stop and no automatic rearm. Update tests and instructions together.

Evidence: controller/include/R14SeekInterlock.h:21; controller/include/R14SeekInterlock.h:187; G:/My Drive/projects/6dof/pcb/HANDOFF_20261023_R14_SOL7C4_ROUTED_VERIFICATION.md

F05 All-axis alarm command stop is not a live implemented path

P1 | CONFIRMED INTEGRATION GAP

R13 measured LOW at the MCU as healthy/conducting and HIGH for off/open. R14 optos retain that polarity and R30-R35 are now 1.5k 1206 instead of overstressed 220-ohm 0603. Current S3 hardware clears the faulted axis torque latch while retaining healthy latches; alarms do not directly clear the independent U11 common pulse-enable latch. A live bounded-latency firmware alarm producer is therefore essential, and none is present. ALL_TORQUE_ENABLED is logic-Q readback, not torque measurement.

Required closure: Implement and measure all-axis pulse stop and queue flush for one asserted alarm, powered-off drive and disconnected cable. Normalize HIGH as unavailable/fault after confirming actual drive settings; preserve healthy drive torque hold and require explicit recovery. Resolve any requirement for an independent common hardware alarm pulse gate explicitly.

Evidence: controller/main/main.cpp:408; live-SCH.net: U9/U10, U23-U30, U11; G:/My Drive/projects/6dof/pcb/HANDOFF_20261012_R13_BENCH_R14_ROUTING.md

F06 R14 cable and drive-enable contract differs from the working R13 setup

P1 | CONFIRMED INTERFACE CHANGE

All six current DB25 footprints use the corrected face numbering; all 25 local hole positions match the mirrored R13-pad relationship. R13-only crossover adapters must not be reused on R14. R14 adds pin6 SRV-ON sinks and pin9 AUX15V; R13 bench Pn003=1 auto-enable does not prove these new external enable paths control the drive. The latest drive gear is also 2, not the older documentation default.

Required closure: Use a separately verified straight R14 cable. Meter molded contacts 1/13/14/25 and every used signal on every port. Qualify the actual drive DI/common/supply and Pn003/Pn052/Pn061 settings, pin6 current, pin9 supply and healthy-hold/fault behavior. Keep R13-only adapters physically labeled and separate.

Evidence: baseline-library-and-connector-certificate.json; interface-contract.csv; R13-to-R14-logical-delta.json

F07 FIELD_VGOOD threshold does not certify the selected remote sensor voltage

P1 | CONFIRMED MARGIN GAP

U18 still uses 100k/15k divider R80/R81 and nominal 1.242V reference: approximately 9.522V local FIELD_12V threshold before tolerances, clamp leakage and hysteresis. Selected NJK-5002D review requires at least 10V at the remote sensor. Even zero cable drop does not make 9.522V proof of 10V. The 15ft cable gauge/contact resistance and loaded supply minimum remain unspecified. Prior feasibility work already identified this gap; current source has not closed it.

Required closure: Freeze the actual cable, sensor load, minimum loaded J9 voltage and temperature envelope. Select/qualify a monitor threshold and power-loss budget that proves the remote minimum, or submit a reviewed monitor/power-path/sensor change. Continue to treat the present field-good bit as a local threshold indication only.

Evidence: hardware/pcb/6dof2_r14/R14_FIELD_VGOOD_15FT_REVIEW_IN_PROGRESS_20261017.md; hardware/pcb/6dof2_r14/R14_FIELD_VGOOD_12V_FEASIBILITY_FOLLOWUP_20261017.md; live-SCH.net: U18,R80-R83,D9

F08 Auxiliary boost and power-sequence guarantees remain conditional

P1 | OPEN HARDWARE QUALIFICATION

AUX15V is nominally 16.4686V from 1.229*(1+124k/10k). The reviewed U37 UV falling lower corner is 12.69874V, only 98.74mV above the assumed 12.6V VIN ceiling. U38 is referenced to eFuse RTN and operates only up to 18V relative RTN; U39 60V rating is not a board rating. Monitor startup, RTN-to-GND displacement, floating logic, retained AUX charge/backfeed and supply maximum are not fully bounded. No actual self-enable counterexample was found in this review.

Required closure: Run the pinned cold/USB-first/VIN-first/drop/reconnect/retained-charge qualification with RTN-aware probing. Establish real VIN/load/temperature/capacitance limits and demonstrate no spontaneous rearm or unintended pulses. Preserve existing separate pulse and torque-latch responsibilities.

Evidence: BOOST_TRIAL_SOL7_20261023/STARTUP_FAULT_QUALIFICATION_20261023/INDEPENDENT_BOOST/INDEPENDENT_C4_BOOST_QUALIFICATION_REVIEW.md; BOOST_TRIAL_SOL7_20261023/STARTUP_FAULT_QUALIFICATION_20261023/INDEPENDENT_POWER/C4_POWER_PATH_STARTUP_REFERENCE_REVIEW_20261023.md

F09 Assembler process approval is still required for reduced margins

P1 | OPEN ASSEMBLY / DFM QUALIFICATION

Fresh CAM confirms no via-drill intersections with SMT mask or paste, but 57 vias have positive aperture gaps below the 0.125mm review band; minimum mask gap is 0.032061mm. U31/U32 DCK pads have only about 0.030mm nominal toe and 0.025mm side margin against the TI maximum envelope. Four J11 plated shell slots intentionally share their own paste windows. J12 native centroid differs from reference anchor by -1.27mm in Y. These are not shorts or automatically wrong parts.

Required closure: Obtain assembler acceptance of stencil/EP/via process, U31/U32 pattern, J11 shell pin-in-paste and exact J12 centroid/rotation. Retain all 11 required non-SMT references. Do not silently rotate connectors or omit required parts to clear order warnings.

Evidence: actual-apertures/APERTURE_VERDICT.json; RESUME_SOL_20261022/U31_U32_DCK_REVIEW_SOL_20261023/U31_U32_DCK_FACT_REVIEW_20261023.md; current-CPL.csv

F10 Collector wrong-port protection and actual mechanical mating remain open

P1 | OPEN PHYSICAL INTERFACE QUALIFICATION

J10 is an integrated Ethernet MagJack; J12 is a plain field jack with power on contact2. Both use 8P8C mating. A powered field cable can be plugged into Ethernet unless the actual enclosure/cable retention prevents it. The field eFuse does not prove W5500 wrong-port survival. Generic connector models do not certify hoods, fasteners, sensor fit or full mating clearance.

Required closure: Prove captive/keyed/noninterchangeable cable handling at both field cable ends and actual enclosure fit. Verify the installed NC sensors and straight-through collector wiring before field power.

Evidence: hardware/pcb/6dof2_r14/FIELD_INTERFACE_CONTRACT.md; hardware/pcb/6dof2_r14/SENSOR_IO_SELECTION.md; live-SCH.net: J10,J12

F11 USB host-line-state and GPIO readback fixes are not in this repo

P2 | CONFIRMED MISSING BENCH CHANGES

The October20 DTR handoff specifies DTR=True, RTS=False and INPUT_OUTPUT for GPIO47/48 readback. Current r13bench_core.py and cobs_console.py both set DTR/RTS False; app_main configures GPIO47/48 OUTPUT. The native USB COBS path remains VFS/usb_serial_jtag-driver based. These omissions can reproduce silent fresh-boot console sessions and misleading status, independently of PCB correctness.

Required closure: Recover the exact tested host and firmware changes; verify replies after a fresh flash with no physical RESET and distinguish requested output state from actual readback. Keep any future transport bounded and avoid blocking motion tasks.

Evidence: hardware/pcb/6dof2_r13/bringup/r13bench_core.py:199; controller/tools/cobs_console.py:59; controller/main/main.cpp:2128; G:/My Drive/projects/6dof/pcb/HANDOFF_20261020_R13_USB_CONSOLE_DTR.md

F12 Old docs and source comments can select the wrong baseline

P2 | CONFIRMED DOCUMENTATION DRIFT

Overview docs still say R13 was not ordered/hardware-tested and some alarm comments say polarity is unknown. R13 bench source names J3 MOTOR0 while current R14 silk says AXIS0. Old R13 testpoint numbers differ from current R14 TP7 ENABLE / TP8 STOP. The fresh report uses exact pin nets and latest dated bench evidence; historical reports are not all current instructions.

Required closure: Reconcile docs after recovering the tested firmware. Publish one R14 pin/testpoint/build/drive-settings contract and clearly distinguish historical R13 results, current source, and pending R14 measurements.

Evidence: docs/FIRMWARE.md:5; controller/main/main.cpp:406; every-label.csv

F13 Ethernet/SPI/USB connectivity pass is not complete signal-integrity qualification

P2 | OPEN PERFORMANCE QUALIFICATION

All 51 current high-speed trace projections were evaluated. Eight MDI records cross the intentional MagJack entry/reference voids; the longest individual unsupported centerline is 5.37mm. USB and SPI projections outside transition exclusions are supported. The geometry audit does not certify end-to-end impedance, EMI or the retained Ethernet via exceptions. Firmware SPI clock remains 26MHz on the matching GPIOs.

Required closure: Review the fabricator stack and retained MDI exceptions against the manufacturer layout guidance; test USB and Ethernet under six-axis switching load. Do not treat a uniform-section impedance calculation as proof at discontinuities.

Evidence: actual-cam/ACTUAL_SIGNAL_REFERENCES.json; signal-reference-summary.json; controller/include/helpers.h:83

Section-by-section review

Each section compares the present source and physical connectivity with the stated reference basis. Component ownership in the register is a navigation aid; shared bypass/passive parts may be assigned by proximity. Pin-net records, not section labels, define the electrical connection.

1. 12V input and buck supply

PASS TOPOLOGY; LOAD/THERMAL QUALIFICATION OPEN

Scope: J9, F1, D1/D2, U6, L1, C1-C3 and associated bypass

The current schematic routes J9 positive through the resettable fuse and reverse-feed diode to VINRAW, then the XL1509-5.0E1 buck and 68uH L1 to VCC5. Ground, catch-diode and feedback membership were checked in the live netlist and actual copper. F1 is the selected 1.1A/16V-class part; this is a nominal 12V board, not a generally 24V/60V-tolerant assembly. All routed supply objects participate in the fresh CAM graph.

Remaining qualification: The supply minimum/maximum, simultaneous six-drive/field/indicator load and temperature envelope are not measured here. The field-monitor feasibility work explicitly depends on upstream diode/fuse losses and loaded input voltage. A connectivity pass does not establish buck stability, inductor saturation, regulator temperature or fuse derating. See F07/F08.

Reference basis: XL1509 manufacturer reference circuit; current live-SCH.net; C4 power-path review.

2. Logic power and USB/12V coexistence

PASS TOPOLOGY; SEQUENCE QUALIFICATION OPEN

Scope: D3/D8, U7 LDL1117S33R and logic decoupling

USB VBUS and buck-derived VCC5 feed LOGIC_5V through separate SS34 diodes; U7 supplies VCC3V3. The driver rail remains distinct from the USB-fed logic rail. Source net membership and all physical connections agree. Existing switch/filter/ground copper is unchanged from the reviewed C4 routing, apart from removing the intentionally unpopulated C78 site.

Remaining qualification: The ST output-above-input limit and diode reverse leakage matter during supply removal and retained charge. No arbitrary partial-power guarantee or complete thermal/load budget is inferred. The cited C4 power review distinguishes supported fail-safe inputs from unspecified unpowered current. USB-only programming and no unintended output enable still require measured sequence tests.

Reference basis: ST LDL1117 datasheet; MDD SS34 reference; C4_POWER_PATH_STARTUP_REFERENCE_REVIEW_20261023.md.

3. ESP32-S3 module, boot and reset

PASS DIGITAL PIN CONTRACT

Scope: U2 ESP32-S3-WROOM-1-N8R2, SW1/SW2, J1

The exact current module is WROOM-1-N8R2 with integrated antenna, not the earlier module-header arrangement or an octal-PSRAM substitute. All twelve motion GPIOs, six alarm GPIOs, GPIO21 stop, GPIO47 inhibit, GPIO48 arm, native USB GPIO19/20, Ethernet GPIO35-39 and I2C GPIO41/42 match the current logical contract. The 8MB firmware flash setting matches N8. Boot/reset and UART nets are retained.

Remaining qualification: The quad-PSRAM N8R2 choice matters: swapping to an octal-memory variant can occupy GPIO35-37. Radio and full enclosure/antenna behavior are not established by netlist checks. Firmware still does not contain the current R14 peripheral adapter.

Reference basis: Espressif WROOM-1/1U datasheet v1.8 and physical pin table; helpers.h; interface-contract.csv.

4. Native USB and recovery

PASS CONNECTIVITY / ARTWORK; HOST REGRESSION OPEN

Scope: J11, D6 USBLC6-2SC6, D7, R2/R3, R69/R70

The USB-C duplicated D+/D- contacts, CC pulldowns, protection and 33-ohm MCU series legs were traced to the module pins. Both current USB reference projections and actual copper graph pass outside reviewed transition regions. J1 remains a 3.3V UART service header. There is no newly claimed USB-to-UART bridge behind this connector.

Remaining qualification: The current host tools still deassert DTR and RTS, unlike the October20 tested handoff. The COBS firmware path still uses the native USB driver/VFS. F11 covers the missing host/readback changes. J11 shell-slot paste is a separate process matter, not a signal short.

Reference basis: Espressif USB Serial/JTAG hardware/driver guidance; ST USBLC6 reference; October20 DTR handoff; fresh CAM.

5. Ethernet and magnetics

PASS NET/PACKAGE CONTRACT; PERFORMANCE QUALIFICATION OPEN

Scope: U8 W5500, J10 HR961160C, Y1, L2, R42-R45 and analogue support

The current J10 is the HR961160C integrated MagJack. Its transformer/center-tap/LED contacts and W5500 transmit/receive nets were reviewed against the preserved manufacturer reference work. There is no unreviewed extra external transformer block. The 25MHz crystal, analogue supply filtering and termination identities are retained. Firmware SPI GPIO35/36/37/38 with interrupt39 and 26MHz clock match the board; W5500 reset remains board RC, matching reset_gpio_num=-1.

Remaining qualification: Eight of 51 critical source-trace projection records cross the intentional MagJack entry/reference voids. They are explicitly listed in signal-reference-summary.json; the longest individual unsupported centerline is 5.37mm. This is not an unreported connectivity defect. Full channel SI/EMI, actual stack impedance and operation under simultaneous motion traffic remain F13 qualification work.

Reference basis: WIZnet W5500 v1.1.0; HanRun HR961160C official drawing; current-CAM actual signal reference audit.

6. STEP/DIR line drivers and common pulse permission

PASS HARDWARE CONTRACT; LIVE FIRMWARE HOLD

Scope: U3/U4/U5 AM26LS31, U11-U14, series resistors/pulldowns

All six STEP and DIR signals route through their selected 33-ohm series resistors into the correct AM26LS31 inputs and complementary outputs. Driver supply is VCC5; common enables are DRIVE_ENABLE and LOGIC_5V on the complementary enable. The separate U11 latch, stop/inhibit logic and 5V supervisor remain present. All twelve differential output mappings are verified at the correct DB25 logical pads.

Remaining qualification: The checked current GPIO and latch maps preserve the R13 electrical contract; they do not preserve the tested R13 pulse generator code. F01 is a confirmed missing implementation. Common permission must remain inhibited until an explicit valid arm sequence, and pad-level pulse width/count/last-step/stop/DIR guard must be remeasured on the recovered firmware.

Reference basis: TI AM26LS31, SN74LVC1G74/1G98 and TPS3890 references; live netlist; helpers.h; October20 pulse-engine handoff.

7. Drive alarm inputs

PASS CORRECTED RESISTOR/NET CONTRACT; GLOBAL STOP OPEN

Scope: U9/U10, R30-R41, six alarm nets

The six current alarm feed resistors are AC1206FR-071K5L, 1.5k in 1206. This preserves the R13 continuous-current correction: the old 220-ohm 0603 overload is not carried forward. At 12.6V with a 1.2V LED assumption, the simple nominal screen is 7.6mA and 0.0866W per resistor; real tolerance, saturation and thermal derating still apply. Healthy-conducting LOW and open/off HIGH agree with R13 measurements and the current pullups.

Remaining qualification: The installed drive variant and provoked-alarm state still require complete cable/polarity tests. A one-axis S3 latch clear is not an implemented all-six command stop. The live alarm adapter/callback connection and measured latency are absent; see F05.

Reference basis: AASD manual and measured R13 healthy-conducting result; ALARM_INPUT_RESISTOR_REVIEW; actual R30-R35 MPNs.

8. Corrected DB25 interfaces and cabling

PASS NUMBERING CORRECTION; PHYSICAL CONTACT GATE OPEN

Scope: J3-J8, current vertical face-numbered footprint

The 150 connector pads use the corrected vertical-female footprint. Each current pad N lies at the corresponding mirrored legacy R13 hole, with no unrelated connector pose change. Pin3=STEP+,14=STEP-,4=DIR+,5=DIR-,10=GND,23=alarm. R14 additionally uses pin6 for SRV-ON and pin9 for auxiliary supply. All six connectors share the same corrected library geometry. Visible 1/13 digits were checked in actual CAM.

Remaining qualification: The actual molded socket contacts, cable continuity, drive DI common and SRV-ON settings are not verified by CAD numbering alone. R13-only crossover adapters are incompatible with the corrected R14 map. The earlier right-angle-origin explanation was corrected by an erratum; the measured mirrored numbering, not that explanation, is the authority. See F06.

Reference basis: CONNFLY DS1034 RevH; October16 physical defect/correction handoffs; baseline-library-and-connector-certificate.json.

9. I2C I/O, reset and initialization

PASS PIN ALLOCATION; FIRMWARE MISSING

Scope: U15 MCP23017, U16 TPS3839, GPIO41/42, pullups

MCP23017 address pins are low for address0x20; INA226 shares the bus at 0x40. Hall inputs are GPA0-5, field-good GPA6, release/clear GPA7. GPB1 controls field feed, GPB2/3 are spare/local-hold nets, GPB4 reads field fault, GPB5 reads DRIVE_ENABLE, GPB6 reads the six-Q aggregate, GPB7 supplies the arm clock. GPA7 and GPB7 are used as outputs, respecting the current Microchip restriction. Both expander interrupt outputs are unused.

Remaining qualification: The current repo has no implemented full-size R14 I2C adapter. Cold-start OLAT/direction ordering, readback, stale-bus detection, warm-reset reconciliation and measured polling latency must be implemented. Do not infer cold POR after an MCU-only reset. See F02/F04/F05.

Reference basis: Microchip DS20001952; TI TPS3839; exact live U15 pin map; current firmware search and admission code.

10. Protected field supply and voltage-good

PASS WIRING; KNOWN SENSOR-SUPPLY MARGIN GAP

Scope: U17 TPS26621, U18 TLV3012B, R74-R83, D9

The field eFuse has the selected26.7k current-limit resistor,1.2M UVLO bypass connection and a ground-reference separation between RTN and board GND. Its output feeds FIELD_12V and J12 contact2. U18 compares the clamped100k/15k divided field rail to its1.242V reference. R82 is a10k series input resistor; D9 is the specified clamp. The current nominal threshold is9.522V.

Remaining qualification: That local9.522V threshold cannot prove the selected sensor10V minimum at the end of a15ft cable. Actual cable DCR, contacts, load, supply droop and tolerances are required. This is an already identified unresolved design margin, not closed by the eFuse rating. See F07.

Reference basis: TI TPS2662/TLV3012B; FIELD_FEED_PROTECTION_REVIEW; October17 field-voltage feasibility records.

11. Hall input conditioning and homing interface

PASS SIX-CHANNEL TOPOLOGY; FIELD QUALIFICATION OPEN

Scope: J12, U19/U20, R84-R95, C48-C53, D10-D15

J12 is the plain RJHSE-5080 field connector:1=GND,2=FIELD_12V,3-8=Hall0-5. Each field signal uses its selected1.1k1210 LED-feed resistor, optocoupler channel and conditioned22k pullup/1nF input filter. Every Hall pad membership and all six U15 inputs match the source. Six independent states are retained; no opposite-end sensors or18-state capability is invented.

Remaining qualification: The selected NC sensors must actively sink away from the magnets. Open/unpowered signals and a sensor at its magnet can look alike; a stuck-low conductor can pass static presence. The source/input filter time constant is about 22us before opto, polling and debounce effects. Bounded seek/time/count and actual cold-bottom-sector measurements remain mandatory commissioning inputs; current code does not implement the live path.

Reference basis: Historical collector contact map; OMCH NJK-5002D selection; Hall conditioning/seek design notes and current netlist.

12. Hall indicators and valid-supply blanking

PASS LOGIC/NET CONTRACT; PARTIAL-SUPPLY INDICATION CONDITIONAL

Scope: U21 AHCT14, D16-D27, Q1-Q3 and LED resistors

The six red/green pairs are driven from conditioned Hall sense nodes through the 5V AHCT14; they do not consume the expander Hall ports or add loads directly to the raw collector signal. The selected1.5k LED resistors, reset-dependent green-return/red-supply switching and exact transistor pin memberships remain present. Source color-part identities are retained.

Remaining qualification: AHCT valid operation is specified at 4.5-5.5V; neither color is an unconditional magnetic-home or safety verdict during power collapse. The reset/blanking circuit is indication logic. Its rail ramp behavior still belongs in power-sequence tests; it does not substitute for the MCU admission state machine.

Reference basis: TI SN74AHCT14; HALL_INDICATOR_DESIGN and HALL_INDICATOR_BLANKING_REVIEW; C4 power review.

13. Six stored torque enables

PASS SOURCE LOGIC CONTRACT; ENABLE/RECOVERY QUALIFICATION OPEN

Scope: U23-U35, Q5-Q10, R113-R132, ALL_TORQUE_ENABLED

Each latch has D and preset tied high; common arm clock is U15GPB7. Individual active-HIGH alarm or global clear drives its active-LOW clear. Global clear includes expander reset/release and auxiliary BAD. Q drives the corresponding MMBT3904 sink through2.7k, with 100k base pulldown and fitted0-ohm connector link. The two three-input ANDs plus final AND report six logicQ states.

Remaining qualification: An ordinary pulse stop need not drop healthy torque. An individual alarm clears its own torque latch while firmware must stop all commanded motion. Expander state can persist through MCU reset; boot must not replay an old arm edge. LogicQ aggregate is not real torque feedback, and Pn003=1 auto-enable from R13 is not proof of externally controlled R14 enable. See F02/F05/F06.

Reference basis: TI LVC1G74, HCS02, LVC1G11/1G08/1G132; S3 source contract and C4 latch startup review.

14. Auxiliary boost and voltage window

PASS CURRENT PIN/REFERENCE TOPOLOGY; POWER ENVELOPE OPEN

Scope: U36-U40, U39 eFuse, L3, D30/D31, R133-R151, C73-C82

TPS61170 feedback is124k/10k, so the AUX15V net is16.4686V nominal. U37 under-voltage output and U38 joined open-drain logic/over-voltage outputs combine in U40 to make BAD. U38 ground and U39 control are RTN-referenced; U37/U40 are board-GND-referenced. C78 was an intentionally unpopulated optional sense filter and is now removed from both masters; the intended fitted circuit is retained.

Remaining qualification: The source matches the reviewed TI-style boost/window/eFuse topology. It does not establish arbitrary VIN/RTN/startup/backfeed behavior. The small UV/pass-through margin,18V U38 operating limit, startup unknown interval and retained-charge decay remain F08. The manufacturer model packages previously inspected were not a worst-case full-temperature switching simulation.

Reference basis: TI TPS61170, TPS2662, TPS3700, TLV6700; exact C4 boost net/DC/corner review and power/latch startup reviews.

15. 5V monitoring and enable indication

PASS HARDWARE CONTRACT; FIRMWARE MISSING

Scope: U22 INA226, R112/C57, Q4/D28, U15 readback

INA226 VBUS measures filtered VCC5 through100ohm R112 and the local bypass/filter. IN+ and IN- are both grounded: this installation measures bus voltage, not load current. DRIVE_ENABLE is routed to U15 for logic readback and to the output indicator path. The selected INA226 permits the bus voltage to be present independently of its supply under the cited reference conditions.

Remaining qualification: Firmware currently does not use these R14 diagnostics. Do not display calculated motor current from this wiring, and do not confuse a logic-enable LED/bit with six actual drive torque states. See F02/F11.

Reference basis: TI INA226; R13 readback enhancement handoff; current U22/U15/Q4 net membership.

16. Copper, return paths, outline and drilling

PASS EXHAUSTIVE DIGITAL CONNECTIVITY / GEOMETRY

Scope: All 1724 native copper traces,878 vias, six pours, fills and test pads

Fresh FS3.6 Gerbers were independently polygonized with dark/clear polarity and actual drilled holes. Regions were joined between layers only through real plated contacts; ground was not joined by a blanket software assumption. All 239 named physical nets form one pad-bearing group each;1062 named physical pads contact actual copper. No missing pads, foreign shorts, unintended named/NC contacts, floating copper,6mil-minus-2um foreign-gap findings, copper-edge findings, keepout findings or NPTH copper-gap findings were detected. All 341 ground through-objects join the actual main Inner1 plane.

Remaining qualification: Every source copper object is indexed in every-trace-via-fill.csv, including 1724 traces and878 vias; all 2621 trace/via/fill/test-pad rows have actual-copper attribution.Drilling covers 1073 unique operations and575128 pairs, minimum analytic edge gap0.3027566mm. Numeric geometry does not prove ampacity, EMC or semiconductor behavior. Critical MDI reference exceptions are explicitly retained in F13.

Reference basis: Current Gerber/current archive; actual-cam outputs; actual-drills report; per-primitive coverage register.

17. Silkscreen, labels and service/test points

PASS TEXT IDENTITY / ACTUAL INK GEOMETRY

Scope: 319 designators,24 board strings,12 fixed DB25 digits

All 319 current component designators are present exactly once and match their owners. Twenty-four board strings retain AXIS0-5, NETWORK, HALL COLLECTOR, USB,12V IN,R14 and the repository URL. Twelve DB25 1/13 digits match the corrected footprint and were visually read in the actual-CAM montage. Whole top ink has zero overlap with mask openings and zero off-board area under the documented1um edge allowance. Every label has a source/pose/ink-region entry.

Remaining qualification: Current testpoints are TP1/2GND,TP3VINRAW,TP4VCC5,TP5LOGIC_5V,TP6VCC3V3,TP7DRIVE_ENABLE,TP8ESTOP_MCU,TP9M0S_I,TP10M0D_I and TP11AUX15V_CTRL. Old R13 TP8-enable instructions do not apply. TP11 has only its reference printed; its function must stay in the current testpoint table. Rendering is a digital artwork witness, not a real socket digit/assembly-placement test.

Reference basis: every-label.csv; current GTO/GTS/GKO; full-silk review; eight current artwork tiles; native standalonePAD records.

18. Assembly identities and mechanical interface

PASS BOM/CPL IDENTITY; PROCESS / FIT OPEN

Scope: 319 required parts,90 supplier codes,306 library documents

Fresh default native BOM has90 rows for90 unique supplier codes and319 distinct references. Fresh CPL has exactly the same319 references; C78 is absent. All 319 schematic/PCB purchasing identities and pin memberships agree. The306 Device/Symbol/Footprint documents are unchanged from the reviewed C4 library set. All 319 retained parts have source-assigned3D models. The current archive has no non-header record change from the saved grouped master.

Remaining qualification: This run did not renew every supplier allocation, obtain factory stencil/rotation approval, export a new assembled STEP, or measure real hoods/cables/enclosure. The old320-source STEP still includesC78 and is not a current manufacturing assembly artifact. U31/U32 margins,J11 shell paste,J12 centroid and11 non-SMT processes remain F09. Field/network wrong-port and physical fit remain F10.

Reference basis: current-BOM/CPL; archive-successor-delta.json; baseline-library-and-connector-certificate.json; assembly and mechanical source reviews.

19. Current repository and working R13 baseline

FAIL COMPATIBILITY / FEATURE COMPLETENESS

Scope: Full-size controller, shared core, host tools and bench handoffs

Current branch is r13-single-motor-sim-20261011 at commit7cc29d4 (full hash pinned), with shared-core commit118e6b0965cee674994bd66d7b9fb7f07c7420e8 and pre-existing uncommitted documentation/app changes. The latest working physical baseline is the October20 bench log, including on-controller waveform runs, pad counter checks and camera return checks at gear2. The log explicitly retracts an earlier host-streamed heave result that computed targets but did not move. That retracted run is not used as success evidence.

Remaining qualification: The tested engine, gear/limit/persistence, fixed tick, hostDTR, readback and expanded bench functions are not present in this checkout. No current live R14 adapter/homing sequence exists. A default fail-closed build and a one-axis bench build both compile;79C++ tests,9 source-contract tests and10 bench-host tests pass. None executes the real MCPWM pulse path or qualifies six-axis R14 hardware. See F01-F05/F11.

Reference basis: Current code snapshot; source hashes; clean review build logs; October20 bench log and two handoffs; firmware-source-manifest.json.

20. Release decision and closure sequence

HOLD / DO NOT ORDER AS A PRODUCTION-READY SYSTEM

Scope: Board, firmware, factory and installed system

Digital hardware checks are strong and reproducible, but loading in JLC and zero native DRC do not close the confirmed software-baseline regressions or the documented field/power/process qualifications. No PCB or firmware source was changed in this review. No image was flashed, motor commanded, order placed, desktop controlled, commit made or push made. The owner working R13 bench behavior has been respected as the required baseline, not overwritten by old repo assumptions.

Remaining qualification: First recover/freeze the exact tested R13 working tree. Then implement the current R14 I/O/enable/home/alarm policy against that baseline and repeat meaningful regressions. Close the field-voltage and power-sequence envelopes, assembler process questions and actual connector/cable/sensor checks. A guarded prototype decision may separately accept named remaining risks; that is not a blanket production PASS.

Reference basis: findings.json; test logs; complete audit registers; reference and hash manifests.

R13 behavior preservation

Feature	Latest working evidence	Current repository / R14 consequence
Pulse production	TEP-latched force gates; direct IRQ; phase accounting; PCNT/camera checks	Old immediate-compare/callback structures retained. F01 blocks equivalence.
Scale and limits	Pn098=2 / Pn102=1; 694.4445 pulses/degree; 50,000-step 72-degree bound	virtual_gear=80 and fixed +/-100,000 steps. Recover tested validation/persistence/limit recalculation.
Tick and bench functions	Fixed 4 us tick; expanded on-controller wave/heave and high-rate bench tests	Current bench accepts 4-100 us and has older 5 kHz/10,000-count caps. Do not equate builds.
USB console/readback	DTR asserted, RTS false; GPIO47/48 INPUT_OUTPUT	Both host tools deassert DTR; output-only GPIO setup remains. F11.
DB25 numbering	R13 physical mirror defect handled with R13-only adapters	R14 corrected numbering. Straight verified R14 cable required; old adapter must not be reused.
Alarm resistors	R13 correction to continuous-current-capable feed	R14 preserves 1.5k 1206 R30-R35. Healthy LOW retained; live global stop absent.
Enable/hold	R13 auto-enable proved one-axis bench motion	R14 adds six external SRV-ON latches. Pn003=1 does not validate this feature.
Owner motion policy	Explicit arm/home; no physical hold-to-run; healthy torque retained on ordinary stop	SeekInterlock still expects local_hold and a fresh edge. Revise contract and tests.
R14 diagnostics	New Hall, field-good, ADC and latch hardware	No live MCP23017/INA226/arm/home adapter. Normal admission deliberately false.

The missing working tree must be recovered, frozen and reviewed before porting. Reimplementing the concept from prose would create a new implementation and would not preserve the measured R13 binary/source baseline.

Current R14 testpoint contract

Point	Actual net	Meaning
TP1 / TP2	GND	Board ground
TP3	VINRAW	After input protection
TP4	VCC5	Driver/buck 5 V
TP5	LOGIC_5V	USB/buck ORed logic rail
TP6	VCC3V3	MCU/logic 3.3 V
TP7	DRIVE_ENABLE	Common pulse-enable logic
TP8	ESTOP_MCU	MCU stop net
TP9	M0S_I	Axis-0 STEP driver input
TP10	M0D_I	Axis-0 DIR driver input
TP11	AUX15V_CTRL	Auxiliary control; only reference printed

Old R13 TP8-enable instructions are superseded for this board. Use actual net names when probing.

Validation and limits

Check	Measured result	Practical limit
Native EasyEDA DRC	Zero findings	Not a schematic correctness or transient proof.
Fresh CAM net graph	239 named physical nets / 1,062 named physical pads connected	Logical pins and physical exposed pads are different counts.
Copper coverage	2,621 primitive rows attributed; no unmapped source objects	1,724 tracks + 878 vias + fills/test pads; pours checked separately.
Ground return	341 through objects join actual main Inner1	No blanket virtual ground union used.
Foreign gaps / NPTH / edge	No findings at documented gates	6 mil minus 2 um; NPTH 0.254 mm minus 2 um; edge 0.3 mm.
Drilling	1,073 unique holes; 575,128 pairs; minimum 0.3027566 mm	1,069 circles + 4 slots; via-only output duplicates excluded.
SMT apertures	1,043 unique pads; 878 vias; zero drill intersections	57 reduced positive gaps need process approval.
Silk/outline	Zero mask overlap/off-board ink; one closed outline	Digit and eight-tile visual inspection complete; actual mating still open.
Host C++ tests	79 / 79 PASS	Host logic, not real ISR execution.
Source-contract tests	9 / 9 PASS	Static source expectations, not physical behavior.
R13 bench-host tests	10 / 10 PASS	Host console/core logic, not a new motor test.
ESP-IDF builds	Standard fail-closed + one-axis bench: exit 0	IDF 5.5; neither image flashed; normal admission remains blocked.

Fresh builds were isolated from the source tree. The firmware manifest freezes 72 source files plus shared-core/test/build inputs. The source-contract tests and build success do not exercise the missing tested pulse-engine replacement or demonstrate six-axis firmware feature completeness.

Electrical and assembly limits

No new full switching simulation or measured six-axis load/thermal envelope was produced. The 0.125 mm aperture review band is a review screen, not an asserted factory minimum. Positive ~32 um gaps and narrow U31/U32 toes are process questions, not detected shorts. Earlier audit helper prose can retain predecessor part counts; current measured count fields and the 319-part coverage certificate govern this review.

Reference and evidence index

Primary reference documents

ESP32-S3-WROOM-1 / 1U, v1.8
https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.html

MCP23017, DS20001952
https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP23017-Data-Sheet-DS20001952.pdf

AM26LS31
https://www.ti.com/lit/ds/symlink/am26ls31.pdf

W5500, v1.1.0
https://docs.wiznet.io/img/products/w5500/W5500_ds_v110e.pdf

TPS2662 family
https://www.ti.com/lit/ds/symlink/tps2662.pdf

TPS3700
https://www.ti.com/lit/ds/symlink/tps3700.pdf

TPS61170
https://www.ti.com/lit/ds/symlink/tps61170.pdf

TLV6700
https://www.ti.com/lit/ds/symlink/tlv6700.pdf

TLV3012B
https://www.ti.com/lit/ds/symlink/tlv3012b.pdf

INA226
https://www.ti.com/lit/ds/symlink/ina226.pdf

LDL1117
https://www.st.com/resource/en/datasheet/ldl1117.pdf

XL1509
https://www.xlsemi.com/datasheet/XL1509%20datasheet-English.pdf

SN74LVC1G11 / DCK pattern
https://www.ti.com/lit/ds/symlink/sn74lvc1g11.pdf

SN74LVC1G74
https://www.ti.com/lit/ds/symlink/sn74lvc1g74.pdf

The component register supplies an individual reference basis for every part. Manufacturer package drawings and prior local reference-circuit reviews are the source for retained connector, optocoupler, indicator and passive choices. No third-party marketing page substitutes for a manufacturer pin map.

Local source and dated bench evidence

G:/My Drive/projects/6dof/pcb/HANDOFF_20261020_STEP_ENGINE_PULSE_ACCOUNTING.md

G:/My Drive/projects/6dof/pcb/HANDOFF_20261020_R13_USB_CONSOLE_DTR.md

G:/My Drive/projects/6dof/pcb/r13-bringup-notes-20261012/BRINGUP_LOG.md

G:/My Drive/projects/6dof/pcb/HANDOFF_20261023_R14_SOL7C4_ROUTED_VERIFICATION.md

G:/My Drive/projects/6dof/pcb/HANDOFF_20261016_R13_DB25_FOOTPRINT_DEFECT.md

Prior electrical reviews live under revisions/2026-10-20-r14-power-first/RESUME_SOL_20261022, especially BOOST_TRIAL_SOL7_20261023/STARTUP_FAULT_QUALIFICATION_20261023 and U31_U32_DCK_REVIEW_SOL_20261023. Repository hardware/pcb/6dof2_r14 contains the field-interface/sensor/monitor feasibility contracts. Relative source citations in findings use this repository or review directory.

Machine-readable coverage files

File	Coverage
every-component.csv	319 identities, poses, footprints, all pin nets and dispositions
every-pin.csv	1,195 logical pin entries
every-trace-via-fill.csv	2,621 source primitive entries and actual net attribution
every-label.csv	355 labels/digits with source/pose/ink entry
interface-contract.csv	78 GPIO/DB25 pin-contract checks
findings.csv / findings.json	13 finding records and closures
actual-cam/	Physical net partitions, copper groups, reference projections
actual-drills/; actual-apertures/; full-silk/	Drill, stencil/mask and full-ink evidence
firmware-source-manifest.json; *tests.log; idf-*-build.log	Frozen source identity, test results and both firmware builds

The compact evidence ZIP includes the PDF, registers, exact native archive/CAM, audit outputs, test logs, review scripts and frozen firmware sources. Installed Python packages, host-build intermediates and review-only binaries are excluded. The SHA256SUMS manifest identifies packaged evidence.

Appendix: every current component

All rows: identity and pin-net agreement PASS. Electrical/assembly disposition remains the corresponding section review and F01-F13; this table is not a per-part unconditional electrical qualification. The CSV retains exact purchasing codes, footprint, pose and full pin mapping.

Ref	Selected manufacturer part / value	Supplier	Section
C1	RVT1E101M0607	C3338	12V input / 5V buck
C2	RST220UF25V019	C4747974	Logic 5V / 3V3 power
C3	CL31A107MQHNNNE	C15008	Logic 5V / 3V3 power
C4	CC0603KRX7R9BB104 / 100nF	C14663	12V input / 5V buck
C5	CC0603KRX7R9BB104 / 100nF	C14663	12V input / 5V buck
C6	CL10A105KB8NNNC / 1uF	C15849	Logic 5V / 3V3 power
C7	CL21A106KAYNNNE	C15850	Logic 5V / 3V3 power
C8	CC0603KRX7R9BB104 / 100nF	C14663	Differential motion outputs
C9	CC0603KRX7R9BB104 / 100nF	C14663	Differential motion outputs
C10	CC0603KRX7R9BB104 / 100nF	C14663	Differential motion outputs
C11	CL21A106KAYNNNE	C15850	Differential motion outputs
C13	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C14	CL10A105KB8NNNC / 1uF	C15849	MCU / boot / reset
C16	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C17	CC0603KRX7R9BB104 / 100nF	C14663	MCU / boot / reset
C18	CC0603KRX7R9BB104 / 100nF	C14663	MCU / boot / reset
C19	CL21A106KAYNNNE	C15850	5V ADC / readback
C20	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C21	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C22	CL21A106KAYNNNE	C15850	Ethernet / magnetics
C25	CC1206KKX7RDBB102	C23631	Hall status indicators
D1	SS34	C8678	12V input / 5V buck
D2	SS34	C8678	12V input / 5V buck
D3	SS34	C8678	USB / recovery
D4	KT-0805G / green	C2297	12V input / 5V buck
D5	KT-0805G / green	C2297	12V input / 5V buck
J1	2.54-1*3P pin	C49257	MCU / boot / reset
J2	KF128-5.08-2P-AA	C474952	Common pulse permission
J9	KF128-5.08-2P-AA	C474952	12V input / 5V buck
J10	HR961160C	C55683	Ethernet / magnetics
J11	TYPE-C-31-M-12	C165948	USB / recovery
L1	CYH125-68UH	C7431083	12V input / 5V buck
R1	0603WAF1002T5E / 10kΩ	C25804	Common pulse permission

Ref	Selected manufacturer part / value	Supplier	Section
R2	0603WAF5101T5E	C23186	USB / recovery
R3	0603WAF5101T5E	C23186	USB / recovery
R4	0603WAF1002T5E / 10kΩ	C25804	MCU / boot / reset
R5	0603WAF1002T5E / 10kΩ	C25804	MCU / boot / reset
R6	0603WAF1001T5E / 1kΩ	C21190	Ethernet / magnetics
R7	0603WAF1001T5E / 1kΩ	C21190	Ethernet / magnetics
R10	0603WAF1242T5E	C22865	Ethernet / magnetics
R11	0603WAF1002T5E / 10kΩ	C25804	Ethernet / magnetics
R12	0603WAF330JT5E	C23140	Differential motion outputs
R13	0603WAF330JT5E	C23140	Differential motion outputs
R14	0603WAF330JT5E	C23140	Differential motion outputs
R15	0603WAF330JT5E	C23140	Differential motion outputs
R16	0603WAF330JT5E	C23140	Differential motion outputs
R17	0603WAF330JT5E	C23140	Differential motion outputs
R18	0603WAF1001T5E / 1kΩ	C21190	12V input / 5V buck
R19	0603WAF1001T5E / 1kΩ	C21190	12V input / 5V buck
R20	0603WAF330JT5E	C23140	Differential motion outputs
R21	0603WAF330JT5E	C23140	Differential motion outputs
R22	0603WAF330JT5E	C23140	Differential motion outputs
R23	0603WAF330JT5E	C23140	Differential motion outputs
R24	0603WAF330JT5E	C23140	Differential motion outputs
R25	0603WAF330JT5E	C23140	Differential motion outputs
U3	AM26LS31CDR	C6179	Differential motion outputs
U4	AM26LS31CDR	C6179	Differential motion outputs
U5	AM26LS31CDR	C6179	Differential motion outputs
U6	XL1509-5.0E1	C61063	12V input / 5V buck
U7	LDL1117S33R	C435835	Logic 5V / 3V3 power
U8	W5500	C32843	Ethernet / magnetics
Y1	X322525MOB4SI	C9006	Ethernet / magnetics
J3	DS1034-25FUNSI44	C77833	Auxiliary boost / fault monitoring
J4	DS1034-25FUNSI44	C77833	Auxiliary boost / fault monitoring
J5	DS1034-25FUNSI44	C77833	Auxiliary boost / fault monitoring
J6	DS1034-25FUNSI44	C77833	Auxiliary boost / fault monitoring
J7	DS1034-25FUNSI44	C77833	Auxiliary boost / fault monitoring
J8	DS1034-25FUNSI44	C77833	Auxiliary boost / fault monitoring
R36	0603WAF1002T5E / 10kΩ	C25804	Drive alarm inputs
R37	0603WAF1002T5E / 10kΩ	C25804	Drive alarm inputs
R38	0603WAF1002T5E / 10kΩ	C25804	Drive alarm inputs

Ref	Selected manufacturer part / value	Supplier	Section
R39	0603WAF1002T5E / 10kΩ	C25804	Drive alarm inputs
R40	0603WAF1002T5E / 10kΩ	C25804	Drive alarm inputs
R41	0603WAF1002T5E / 10kΩ	C25804	Drive alarm inputs
U9	LTV-847S	C114599	Drive alarm inputs
U10	LTV-847S	C114599	Drive alarm inputs
C12	CL10B103KB8NNNC / 10nF	C1589	Ethernet / magnetics
C15	CL21A475KAQNNNE / 4.7uF	C1779	Ethernet / magnetics
C26	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C27	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C28	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C29	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C30	CC0603KRX7R9BB104 / 100nF	C14663	Ethernet / magnetics
C31	CL21A106KAYNNNE	C15850	Ethernet / magnetics
L2	PBY160808T-601Y-N	C108301	Protected Hall field supply
R42	RC0603FR-0749R9L	C114625	Ethernet / magnetics
R43	RC0603FR-0749R9L	C114625	Ethernet / magnetics
R44	RC0603FR-0749R9L	C114625	Ethernet / magnetics
R45	RC0603FR-0749R9L	C114625	Ethernet / magnetics
R58	0603WAF1004T5E	C22935	Ethernet / magnetics
R46	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R47	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R48	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R49	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R50	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R51	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R52	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R53	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R54	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R55	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R56	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R57	0603WAF1001T5E / 1kΩ	C21190	Differential motion outputs
R59	0603WAF100JT5E	C22859	Ethernet / magnetics
C32	CL10B223KB8NNNC	C21122	Ethernet / magnetics
C33	0603B682K500NT / 6.8nF	C1631	Ethernet / magnetics
C34	0603B682K500NT / 6.8nF	C1631	Ethernet / magnetics
C35	CL10B103KB8NNNC / 10nF	C1589	Ethernet / magnetics
D7	SMF5.0A	C193402	USB / recovery
D6	USBLC6-2SC6	C7519	USB / recovery

Ref	Selected manufacturer part / value	Supplier	Section
SW1	TS-1088R-02026	C455280	MCU / boot / reset
SW2	TS-1088R-02026	C455280	MCU / boot / reset
C23	CL10C180JB8NNNC	C1647	Ethernet / magnetics
C24	CL10C180JB8NNNC	C1647	Ethernet / magnetics
F1	BSMD1812-110-16V	C883148	12V input / 5V buck
R60	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
C36	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission
U11	SN74LVC1G74DCUR	C70285	Common pulse permission
U12	SN74LVC1G98DCKR	C485081	Common pulse permission
U13	TPS389001DSER	C1509297	Common pulse permission
U14	SN74LVC1G98DCKR	C485081	Common pulse permission
D8	SS34	C8678	Logic 5V / 3V3 power
R61	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
R62	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
R63	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
R64	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
R65	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
R66	0603WAF1002T5E / 10kΩ	C25804	Common pulse permission
R67	RT0603BRD0732KL	C861325	Common pulse permission
R68	RT0603BRD0710KL	C95204	Common pulse permission
R69	0603WAF330JT5E	C23140	USB / recovery
R70	0603WAF330JT5E	C23140	USB / recovery
C37	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission
C38	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission
C39	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission
C40	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission
C41	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission
C42	CL21A475KAQNNNE / 4.7uF	C1779	Logic 5V / 3V3 power
R30	AC1206FR-071K5L	C229395	Drive alarm inputs
R31	AC1206FR-071K5L	C229395	Drive alarm inputs
R32	AC1206FR-071K5L	C229395	Drive alarm inputs
R33	AC1206FR-071K5L	C229395	Drive alarm inputs
R34	AC1206FR-071K5L	C229395	Drive alarm inputs
R35	AC1206FR-071K5L	C229395	Drive alarm inputs
U15	MCP23017-E/SO	C47023	I2C I/O and initialization
R71	0603WAF4701T5E / 4.7kΩ	C23162	Hall input conditioning
R72	0603WAF4701T5E / 4.7kΩ	C23162	Hall input conditioning
C43	CC0603KRX7R9BB104 / 100nF	C14663	Common pulse permission

Ref	Selected manufacturer part / value	Supplier	Section
U16	TPS3839G33DBZR	C485802	I2C I/O and initialization
R73	0603WAF1003T5E / 100kΩ	C25803	I2C I/O and initialization
C44	CC0603KRX7R9BB104 / 100nF	C14663	I2C I/O and initialization
U17	TPS26621DRCR	C1848341	Protected Hall field supply
R74	RT0603BRD0726K7L / 26.7kΩ	C728587	Protected Hall field supply
R76	RC0603FR-07100KL / 100kΩ	C14675	Protected Hall field supply
R77	0603WAF8201T5E / 8.2kΩ	C25981	Protected Hall field supply
R78	0603WAF1002T5E / 10kΩ	C25804	Protected Hall field supply
R79	0603WAF1002T5E / 10kΩ	C25804	Protected Hall field supply
C45	CL10B105KA8NNNC / 1uF	C29936	12V input / 5V buck
C46	CL10B105KA8NNNC / 1uF	C29936	Protected Hall field supply
U18	TLV3012BIDCKR	C20345925	Protected Hall field supply
R80	RC0603FR-07100KL / 100kΩ	C14675	Protected Hall field supply
R81	RC0603FR-0715KL	C114661	Protected Hall field supply
R82	0603WAF1002T5E / 10kΩ	C25804	Protected Hall field supply
R83	RC0603FR-07100KL / 100kΩ	C14675	Protected Hall field supply
D9	BZT52C3V6	C2113	Protected Hall field supply
C47	CC0603KRX7R9BB104 / 100nF	C14663	Protected Hall field supply
R75	0603WAF1204T5E / 1.2MΩ	C13315	Protected Hall field supply
U19	LTV-847S	C114599	Hall input conditioning
U20	LTV-847S	C114599	Hall input conditioning
R84	AC1210JR-071K1L / 1.1k	C230412	Hall status indicators
R90	0603WAF2202T5E	C31850	Hall input conditioning
D10	1N4148WS	C57759	Hall status indicators
C48	CL10B102KB8NNNC	C1588	Hall input conditioning
R85	AC1210JR-071K1L / 1.1k	C230412	Hall status indicators
R91	0603WAF2202T5E	C31850	Hall input conditioning
D11	1N4148WS	C57759	Hall status indicators
C49	CL10B102KB8NNNC	C1588	Hall input conditioning
R86	AC1210JR-071K1L / 1.1k	C230412	Hall status indicators
R92	0603WAF2202T5E	C31850	Hall input conditioning
D12	1N4148WS	C57759	Hall status indicators
C50	CL10B102KB8NNNC	C1588	Hall input conditioning
R87	AC1210JR-071K1L / 1.1k	C230412	Hall status indicators
R93	0603WAF2202T5E	C31850	Hall input conditioning
D13	1N4148WS	C57759	Hall status indicators
C51	CL10B102KB8NNNC	C1588	Hall input conditioning
R88	AC1210JR-071K1L / 1.1k	C230412	Hall status indicators

Ref	Selected manufacturer part / value	Supplier	Section
R94	0603WAF2202T5E	C31850	Hall input conditioning
D14	1N4148WS	C57759	Hall status indicators
C52	CL10B102KB8NNNC	C1588	Hall input conditioning
R89	AC1210JR-071K1L / 1.1k	C230412	Hall status indicators
R95	0603WAF2202T5E	C31850	Hall input conditioning
D15	1N4148WS	C57759	Hall status indicators
C53	CL10B102KB8NNNC	C1588	Hall input conditioning
U21	SN74AHCT14PWR	C7481	Hall status indicators
Q1	AO3400A / N-MOS	C20917	Hall status indicators
Q2	AO3400A / N-MOS	C20917	Hall status indicators
Q3	AO3401A	C15127	Hall status indicators
R108	0603WAF1003T5E / 100kΩ	C25803	Hall status indicators
C54	CC0603KRX7R9BB104 / 100nF	C14663	Hall status indicators
R96	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D16	LTST-C170KGKT	C98221	Hall status indicators
R102	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D22	LTST-C170KRKT	C94868	Hall status indicators
R97	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D17	LTST-C170KGKT	C98221	Hall status indicators
R103	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D23	LTST-C170KRKT	C94868	Hall status indicators
R98	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D18	LTST-C170KGKT	C98221	Hall status indicators
R104	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D24	LTST-C170KRKT	C94868	Hall status indicators
R99	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D19	LTST-C170KGKT	C98221	Hall status indicators
R105	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D25	LTST-C170KRKT	C94868	Hall status indicators
R100	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D20	LTST-C170KGKT	C98221	Hall status indicators
R106	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D26	LTST-C170KRKT	C94868	Hall status indicators
R101	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D21	LTST-C170KGKT	C98221	Hall status indicators
R107	0603WAF1501T5E / 1.5kΩ	C22843	Hall status indicators
D27	LTST-C170KRKT	C94868	Hall status indicators
J12	RJHSE-5080	C599441	Hall input conditioning

Ref	Selected manufacturer part / value	Supplier	Section
U2	ESP32-S3-WROOM-1-N8R2 / 2.4GHz	C2913204	MCU / boot / reset
R109	0603WAF1001T5E / 1kΩ	C21190	Common pulse permission
R110	0603WAF1003T5E / 100kΩ	C25803	5V ADC / readback
C55	CC0603KRX7R9BB104 / 100nF	C14663	5V ADC / readback
Q4	AO3400A / N-MOS	C20917	Hall status indicators
D28	KT-0805G / green	C2297	Hall status indicators
R111	0603WAF1001T5E / 1kΩ	C21190	Hall status indicators
U22	INA226AIDGSR / I2C rail ADC	C49851	5V ADC / readback
C56	CC0603KRX7R9BB104 / 100nF	C14663	5V ADC / readback
R112	0603WAF1000T5E / 100R	C22775	5V ADC / readback
C57	CC0603KRX7R9BB104 / 100nF	C14663	5V ADC / readback
U23	SN74LVC1G74DCUR	C70285	Six stored torque enables
Q5	MMBT3904(RANGE:100-300)	C20526	Six stored torque enables
R115	0603WAF2701T5E / 2.7kΩ	C13167	Six stored torque enables
R121	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R127	0603WAF0000T5E / 0Ω	C21189	Six stored torque enables
C58	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U24	SN74LVC1G74DCUR	C70285	Six stored torque enables
Q6	MMBT3904(RANGE:100-300)	C20526	Six stored torque enables
R116	0603WAF2701T5E / 2.7kΩ	C13167	Six stored torque enables
R122	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R128	0603WAF0000T5E / 0Ω	C21189	Six stored torque enables
C59	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U25	SN74LVC1G74DCUR	C70285	Six stored torque enables
Q7	MMBT3904(RANGE:100-300)	C20526	Six stored torque enables
R117	0603WAF2701T5E / 2.7kΩ	C13167	Six stored torque enables
R123	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R129	0603WAF0000T5E / 0Ω	C21189	Six stored torque enables
C60	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U26	SN74LVC1G74DCUR	C70285	Six stored torque enables
Q8	MMBT3904(RANGE:100-300)	C20526	Six stored torque enables
R118	0603WAF2701T5E / 2.7kΩ	C13167	Six stored torque enables
R124	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R130	0603WAF0000T5E / 0Ω	C21189	Six stored torque enables
C61	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U27	SN74LVC1G74DCUR	C70285	Six stored torque enables
Q9	MMBT3904(RANGE:100-300)	C20526	Six stored torque enables
R119	0603WAF2701T5E / 2.7kΩ	C13167	Six stored torque enables

Ref	Selected manufacturer part / value	Supplier	Section
R125	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R131	0603WAF0000T5E / 0Ω	C21189	Six stored torque enables
C62	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U28	SN74LVC1G74DCUR	C70285	Six stored torque enables
Q10	MMBT3904(RANGE:100-300)	C20526	Six stored torque enables
R120	0603WAF2701T5E / 2.7kΩ	C13167	Six stored torque enables
R126	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R132	0603WAF0000T5E / 0Ω	C21189	Six stored torque enables
C63	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U29	SN74HCS02BQAR	C4471167	Six stored torque enables
U30	SN74HCS02BQAR	C4471167	Six stored torque enables
C64	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
C65	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U35	SN74LVC1G132DCKR	C205992	Six stored torque enables
C70	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
R113	0603WAF1003T5E / 100kΩ	C25803	Six stored torque enables
R114	0603WAF1002T5E / 10kΩ	C25804	Six stored torque enables
U31	SN74LVC1G11DCKR	C529281	Six stored torque enables
U32	SN74LVC1G11DCKR	C529281	Six stored torque enables
U33	SN74LVC1G08DCKR	C7832	Six stored torque enables
C66	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
C67	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
C68	CC0603KRX7R9BB104 / 100nF	C14663	Six stored torque enables
U39	TPS26621DRCR	C1848341	Auxiliary boost / fault monitoring
R142	RT0603BRD0726K7L / 26.7kΩ	C728587	Auxiliary boost / fault monitoring
R143	0603WAF1204T5E / 1.2MΩ	C13315	Auxiliary boost / fault monitoring
R144	RC0603FR-07100KL / 100kΩ	C14675	Auxiliary boost / fault monitoring
R145	0603WAF8201T5E / 8.2kΩ	C25981	Auxiliary boost / fault monitoring
C79	CC0603KRX7R9BB104 / 100nF	C14663	Auxiliary boost / fault monitoring
C80	CL10B105KA8NNNC / 1uF	C29936	12V input / 5V buck
D30	SS34	C8678	Auxiliary boost / fault monitoring
U36	TPS61170DRVR	C15163	Auxiliary boost / fault monitoring
L3	SWPA5040S100MT / 10uH	C84608	Auxiliary boost / fault monitoring
D31	SS34	C8678	Auxiliary boost / fault monitoring
C71	CL21B105KBFNNNE / 1uF	C28323	Auxiliary boost / fault monitoring
C72	CC0603KRX7R9BB104 / 100nF	C14663	Auxiliary boost / fault monitoring
C73	CL31B106KBHNNNE / 10uF	C89632	Auxiliary boost / fault monitoring
C74	CC0603KRX7R9BB104 / 100nF	C14663	Auxiliary boost / fault monitoring

Ref	Selected manufacturer part / value	Supplier	Section
R133	0603WAF1243T5E / 124kΩ	C22788	Auxiliary boost / fault monitoring
R134	0603WAF1002T5E / 10kΩ	C25804	Auxiliary boost / fault monitoring
R135	0603WAF4991T5E / 4.99kΩ	C23046	Auxiliary boost / fault monitoring
C75	CL10B103KB8NNNC / 10nF	C1589	Auxiliary boost / fault monitoring
R136	0603WAF1003T5E / 100kΩ	C25803	Auxiliary boost / fault monitoring
R138	0603WAF1002T5E / 10kΩ	C25804	Auxiliary boost / fault monitoring
R141	0603WAF1002T5E / 10kΩ	C25804	Auxiliary boost / fault monitoring
C77	CC0603KRX7R9BB104 / 100nF	C14663	Auxiliary boost / fault monitoring
R146	0603WAF1243T5E / 124kΩ	C22788	Auxiliary boost / fault monitoring
C81	CC0603KRX7R9BB104 / 100nF	C14663	Auxiliary boost / fault monitoring
U38	TPS3700DDCR	C33002	Auxiliary boost / fault monitoring
U37	TLV6700DDCR	C2868382	Auxiliary boost / fault monitoring
U40	SN74LVC1G132DCKR	C205992	Auxiliary boost / fault monitoring
R149	0603WAF6341T5E / 6.34kΩ	C23087	Auxiliary boost / fault monitoring
R150	0603WAF1001T5E / 1kΩ	C21190	Auxiliary boost / fault monitoring
R151	0603WAF4701T5E / 4.7kΩ	C23162	Auxiliary boost / fault monitoring
C82	CC0603KRX7R9BB104 / 100nF	C14663	Auxiliary boost / fault monitoring
R137	0603WAF3323T5E / 332kΩ	C23139	Auxiliary boost / fault monitoring
R147	0603WAF2551T5E / 2.55kΩ	C25965	Auxiliary boost / fault monitoring
R148	0603WAF1501T5E / 1.5kΩ	C22843	Auxiliary boost / fault monitoring

End of review. Release remains HOLD until the named firmware-baseline and qualification findings are closed or explicitly accepted for a bounded prototype. No source repair is implied by this report.